LECTURE 9

LAST TIME

LECTURE 8: SUMMARY

Binomial model

Options on tree

THIS TIME

LECTURE 9 OUTLINE

Risk-neutral pricing

Risk-neutral probabilities

Risk-neutral expectations

BINOMIAL MODEL

REPLICATE WITH GENERAL PAYOFFS $X = (X_u, X_d)$

Step I: Write payoffs (X_u, X_d)

Step II: Replicate in states u and d with β bonds and Δ stocks

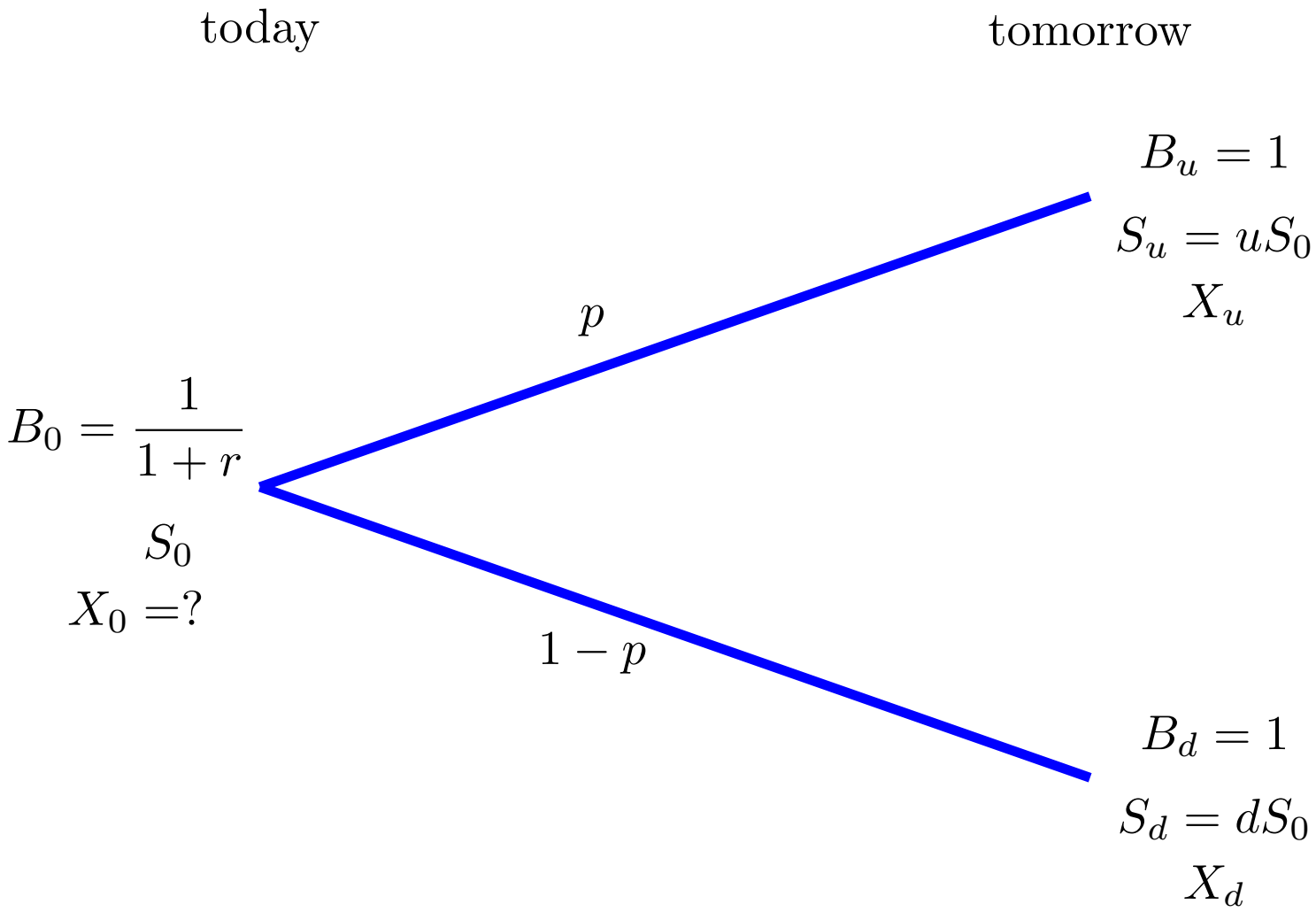
$$X_u = \beta B_u + \Delta S_u$$

$$X_d = \beta B_d + \Delta S_d$$

Step III: Solve system for β Δ

Step IV: By LoOP: $X_0 = \beta B_0 + \Delta S_0$

GENERAL PAYOFFS



PRICE A POWER DERIVATIVE

DERIVATIVES PRICING EXERCISE

Let $S_0 = 3$, $S_u = 5$, $S_d = 2$, $r = 0$, $p = 1/2$

Price a derivative that pays off $X_T = S_T^2$

DERIVATIVES PRICING SOLUTION

Find the replicating portfolio (β, Δ)

I.e.

$$\beta + \Delta S_u = S_u^2$$

and

$$\beta + \Delta S_d = S_d^2$$

or

$$\beta + 5\Delta = 25$$

and

$$\beta + 2\Delta = 4$$

Subtract the above to get

$$3\Delta = 21$$

or $\Delta = 7$

DERIVATIVES PRICING SOLUTION

Hence, from

$$\beta + \Delta S_d = S_d^2,$$

we have

$$\beta = 4 - 2 \cdot 7 = -10$$

Apply the Law of One Price to see

$$\begin{aligned} X_0 &= \beta + \Delta S_0 \\ &= -10 + 7 \cdot 3 \\ &= 11 \end{aligned}$$

Note $11 = X_0 < \mathbb{E}[X_T] = \frac{1}{2} \cdot 25 + \frac{1}{2} \cdot 4 = 14.5$ Comment?

RETURN TO REPLICATING PORTFOLIO AND PRICE OF X

REPLICATE X

Replicate X with portfolio (β, Δ)

$$\beta + \Delta u S_0 = X_u$$
$$\beta + \Delta d S_0 = X_d$$

or

$$\Delta = \frac{X_u - X_d}{(u-d)S_0}$$

and thus

$$\beta = X_d - \frac{(X_u - X_d)d}{u-d}$$
$$= \frac{uX_d - dX_u}{u-d}$$

REPLICATION RECIPE

We computed a way to replicate any derivative X on the tree

Buy

$$\Delta = \frac{X_u - X_d}{S_u - S_d}$$

stocks and

$$\beta = \frac{uX_d - dX_u}{u - d}$$

bonds

This gives a pricing recipe

$$X_0 = \beta B_0 + \Delta S_0$$

DELTA HEDGE

Observe that Δ is sensitivity of X to S:

$$\Delta = \frac{X_u - X_d}{S_u - S_d} \approx \frac{\partial X}{\partial S}$$

DELTA HEDGE QUESTIONS

Suppose X changes a lot for small changes in S

Will the replicating portfolio contain few or many stocks?

How does this relate to leverage?

BACK TO X_0

THE PRICE X_0

Substituting (β, Δ) from above:

$$X_0 \;=\; \beta B_0 + \Delta S_0$$

THE PRICE X_0

Substituting (β, Δ) from above:

$$\begin{aligned} X_0 &= \beta B_0 + \Delta S_0 \\ &= \frac{1}{1+r} \frac{uX_d - dX_u}{u-d} + \frac{X_u - X_d}{u-d} \end{aligned}$$

THE PRICE X_0

Substituting (β, Δ) from above:

$$
\begin{aligned}
X_0 &= \beta B_0 + \Delta S_0 \\
&= \frac{1}{1+r} \frac{uX_d - dX_u}{u-d} + \frac{X_u - X_d}{u-d} \\
&= \frac{1}{1+r} \left[\left(\frac{1+r-d}{u-d} \right) X_u + \left(\frac{u-(1+r)}{u-d} \right) X_d \right]
\end{aligned}
$$

THE PRICE X_0

Substituting (β, Δ) from above:

$$\begin{aligned} X_0 &= \beta B_0 + \Delta S_0 \\ &= \frac{1}{1+r}\,\frac{uX_d - dX_u}{u-d} + \frac{X_u - X_d}{u-d} \\ &= \frac{1}{1+r}\left[\left(\frac{1+r-d}{u-d}\right) X_u + \left(\frac{u-(1+r)}{u-d}\right) X_d\right] \end{aligned}$$

Note:

$$\frac{1+r-d}{u-d} + \frac{u-(1+r)}{u-d} = 1$$

THE PRICE X_0

Substituting (β, Δ) from above:

$$\begin{aligned} X_0 &= \beta B_0 + \Delta S_0 \\ &= \frac{1}{1+r}\,\frac{uX_d - dX_u}{u-d} + \frac{X_u - X_d}{u-d} \\ &= \frac{1}{1+r}\left[\left(\frac{1+r-d}{u-d}\right)X_u + \left(\frac{u-(1+r)}{u-d}\right)X_d\right] \end{aligned}$$

Note:

$$\frac{1+r-d}{u-d} + \frac{u-(1+r)}{u-d} = 1$$

and

$$\frac{1+r-d}{u-d} > 0, \qquad \frac{u-(1+r)}{u-d} > 0$$

THE PRICING FORMULA

Price X_0 depends on

derivative payoffs X_u and X_d

stock returns u and d

bond return/risk-free rate r

Where is probability $p = \mathbb{P}[u]$?

RISK-NEUTRAL PRICING

RISK-NEUTRAL PROBABILITIES

Introduce the notation

$$q \equiv \mathbb{Q}[u] := \frac{1 + r - d}{u - d}$$

to simplify pricing formula to get:

$$X_0 = \frac{1}{1 + r}\Big[qX_u + (1 - q)X_d\Big]$$

RISK-NEUTRAL PRICING EXERCISE

Let $S_0 = 100$, $S_u = 120$, $S_d = 80$, $r = 10\%$, $p = .8$

Find the risk-neutral probabilities

Price a call on S with strike $K = 100$

Price a derivative Y that pays off $Y_u = 80$ and $Y_d = 2$

RISK-NEUTRAL PRICING SOLUTION

First find u and d

$$u = \frac{S_u}{S_0} = \frac{120}{100} = 1.2$$

and

$$d = \frac{S_d}{S_0} = \frac{80}{100} = 0.8$$

Now, find q from the formula

$$\begin{aligned} q &= \frac{1 + r - d}{u - d} \\ &= \frac{1.1 - 0.8}{1.2 - 0.8} \\ &= \frac{3}{4} \end{aligned}$$

and thus $1 - q = 1/4$

RISK-NEUTRAL PRICING SOLUTION

The call pays off 20 in state u and zero in state d

Thus, from the risk-neutral pricing formula,

$$\begin{aligned} c_0 &= \frac{qc_u + (1-q)c_d}{1+r} \\ &= \frac{.75 \cdot 20}{1.1} \\ &\approx 13.63 \end{aligned}$$

RISK-NEUTRAL PRICING SOLUTION

Y pays off 80 in state u and 2 in state d

Thus, from the risk-neutral pricing formula,

$$\begin{aligned} Y_0 &= \frac{qY_u + (1-q)Y_d}{1+r} \\ &= \frac{.75 \cdot 80 + .25 \cdot 2}{1.1} \\ &= 55 \end{aligned}$$

RISK-NEUTRAL EXPECTATIONS

$\mathbb{Q}$ is mathematical probability (numbers in [0,1] adding to 1)

Risk-neutral pricing formula is expectation

$$X_0 = \frac{qX_u + (1-q)X_d}{1+r} = \frac{1}{1+r}\mathbb{E}^{\mathbb{Q}}\big[X_T\big]$$

But the expectation is taken with $\mathbb{Q}$

A BETTER NPV RULE

A kind of NPV formula:

But the real probably $\mathbb{P}$ is replaced by $\mathbb{Q}$

$\mathbb{Q}$ is not just a mathematical probability

it is used to price, based on market preferences

Formula holds beyond the binomial model

NA is equivalent to the existence of a probability $\mathbb{Q}$

Called the fundamental theorem of asset pricing (FTAP)

THE FORWARD PRICE FROM THE FTAP

FORWARD CONTRACTS

Find the forward price with the FTAP

$$\frac{1}{1+r}\mathbb{E}^{\mathbb{Q}}\left[S_1 - F_0(1)\right] = 0$$

or

$$F_0(1) = \mathbb{E}^{\mathbb{Q}}\left[S_1\right]$$

What is $\mathbb{E}^{\mathbb{Q}}\left[S_1\right]$?

Suppose there are no dividends,

$$S_t = \frac{1}{1+r}\mathbb{E}^{\mathbb{Q}}\left[S_1\right]$$

since the price of a claim on S_1 is S_0 by definition

Thus $F_0 = (1+r)S_0$, as we found before

RECAP

LECTURE 9: RECAP

Derivatives pricing recipe on binomial tree

Risk-neutral pricing

- Risk-neutral probabilities
- Risk-neutral expectations

FTAP

FIN